



Yoga!



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Problem

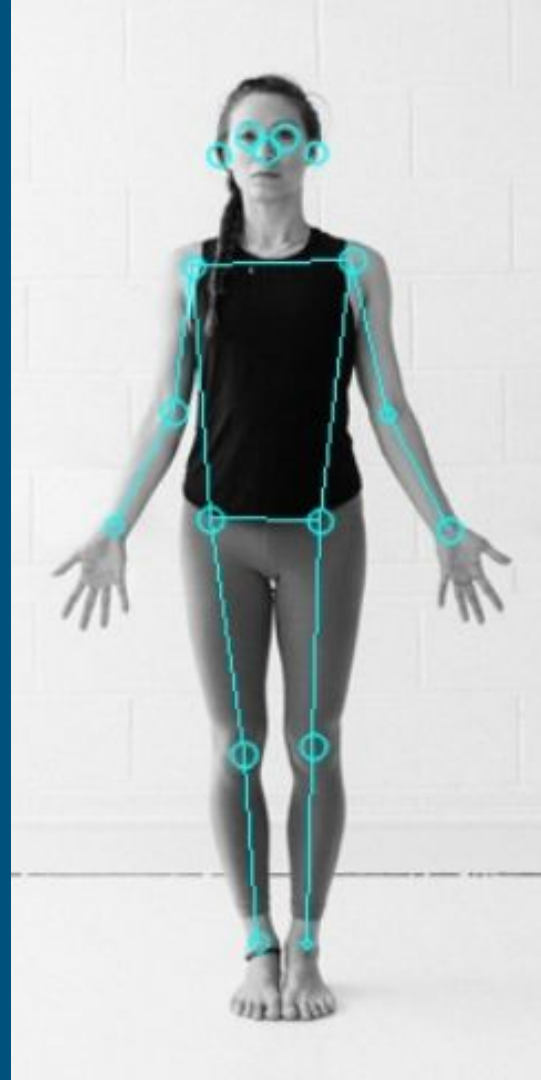
- Big data analytics in athletics focuses on biometrics (pulse, exertion rate) and rate of travel
- Instructions are based on progression of athlete, but not on accuracy of movement
- Risk of injury for athletes whose coaches are remote is higher than necessary



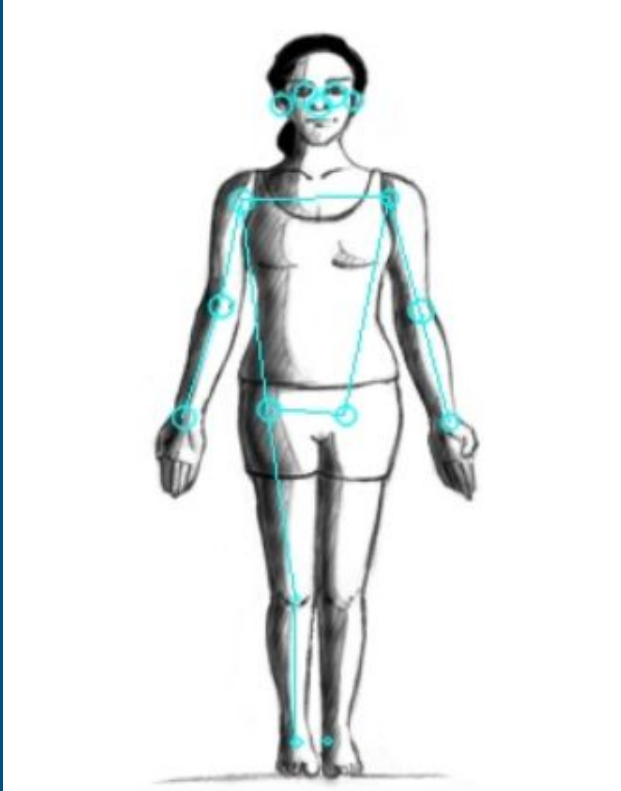
Our Proposed Solution

We've developed YogaPose, pose estimation software that allows pose instructors to teach class at scale and analyze which of their students follow along successfully with recommended courses of instruction.

It's yoga class with an instructor available anytime, anywhere, who can give instantaneous feedback, even without an internet connection, and feedback based on student trends when students are connected.

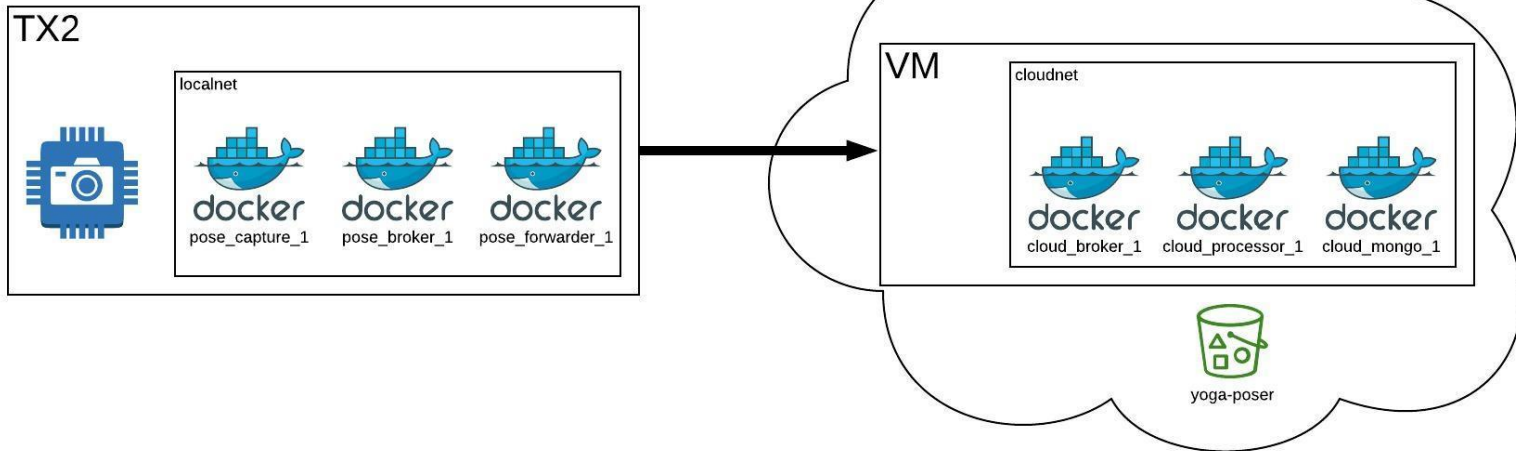


Project Goals



- YogaPose will:
 - Allow a single remote instructor to send out yoga poses to one or more students
 - Require students to strike those poses
 - Send images of students striking poses back to a central server
 - Extract analytics, such as the student's wireframe, and the perceived pose
 - Give the user near real time feedback of pose achieved

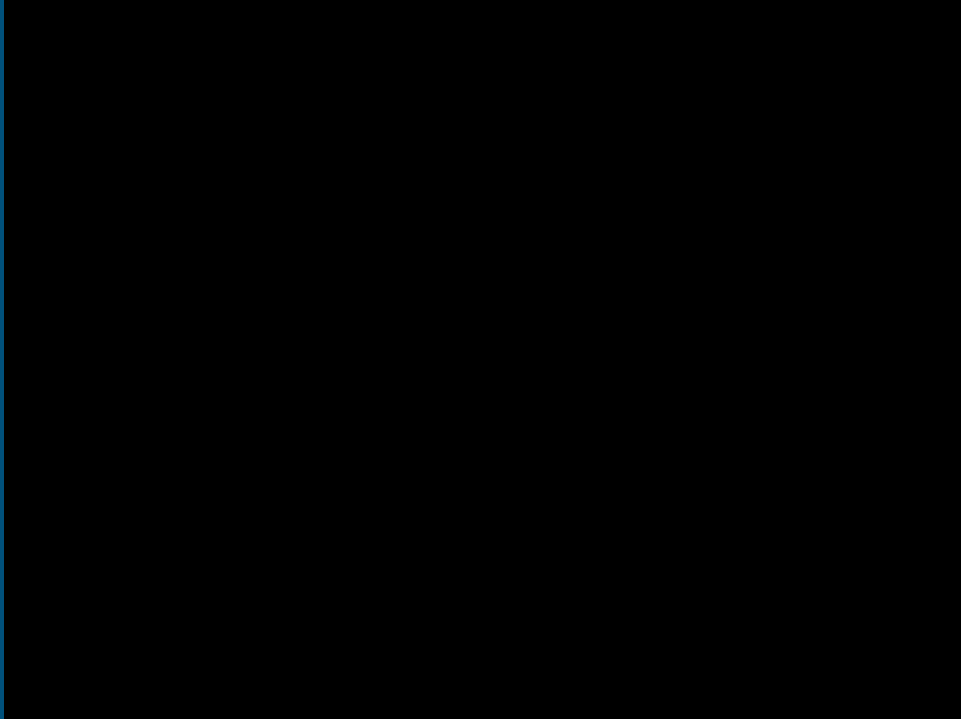
Our proposed architecture



What runs on the TX2

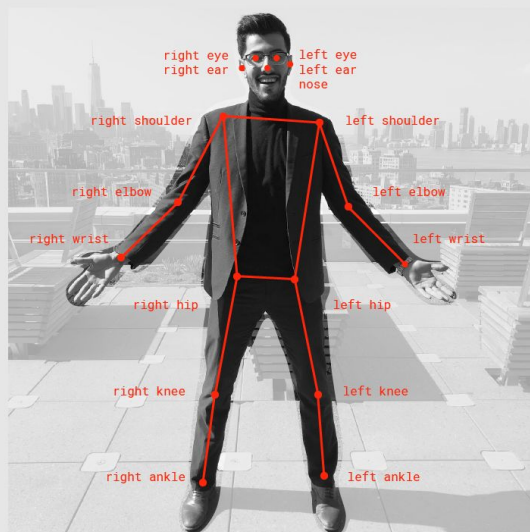
- Image Acquisition
- Pose Estimation and Overlay
- Yoga Pose Classification

Demo



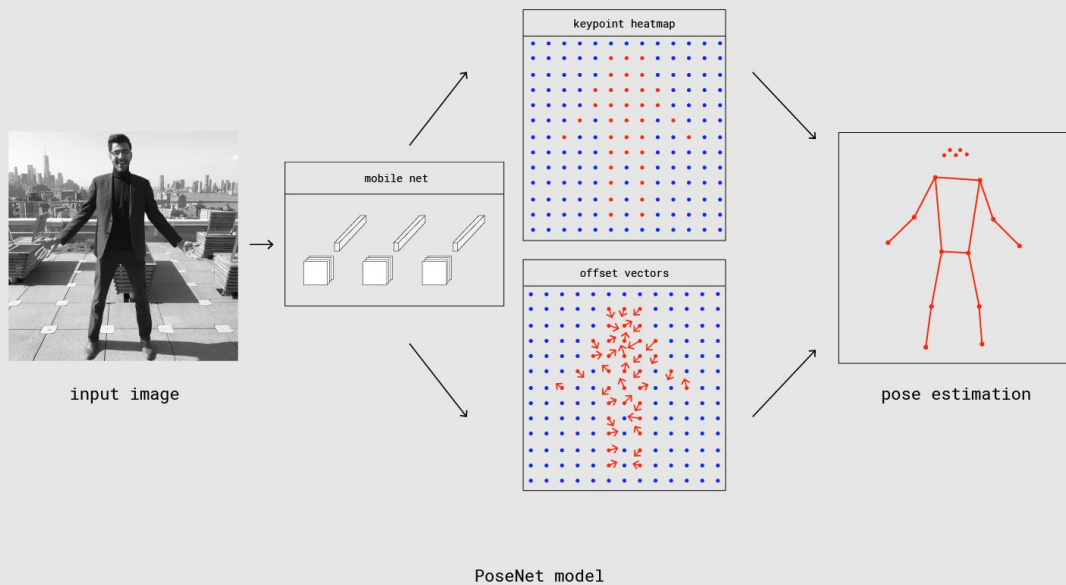
PoseNet

17 Pose Keypoints
Returned by PoseNet



PoseNet

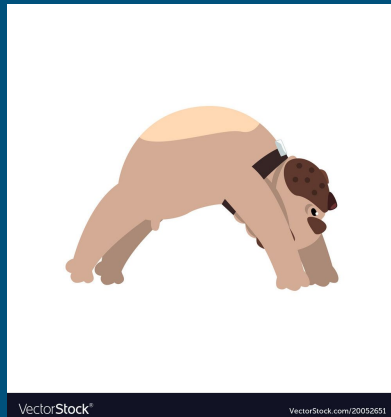
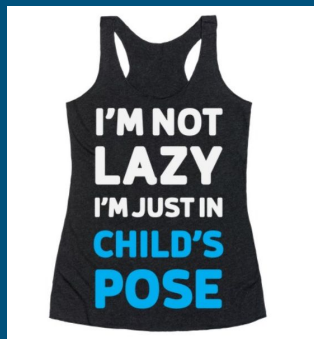
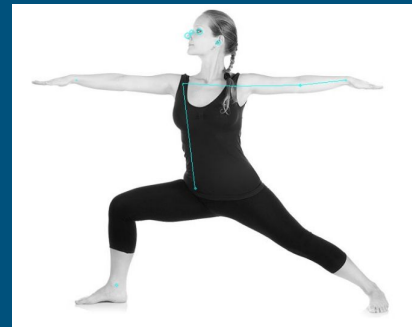
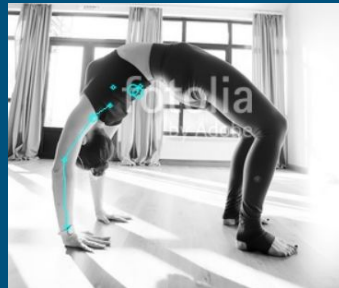
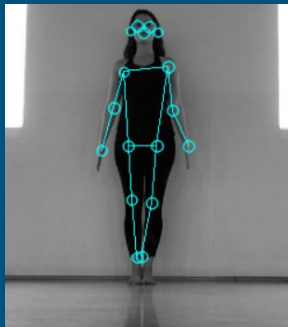
Single-Pose Detection Algorithm



Yoga Pose Classification Dataset

- Google Chrome Driver and Google Image Download 50000 per term with about 2000 downloads to search for key terms resulting in about 1200 images per pose after data was filtered
- PoseNet to filter data
- Resulted in corpus of 6000+ image corpus built from “scratch” in about 4 hours leveraging cloud capabilities (storage capacity and speed)
- Focused on noise reduction to increase model accuracy for PoseNet filter and classification CNN

Image Examples



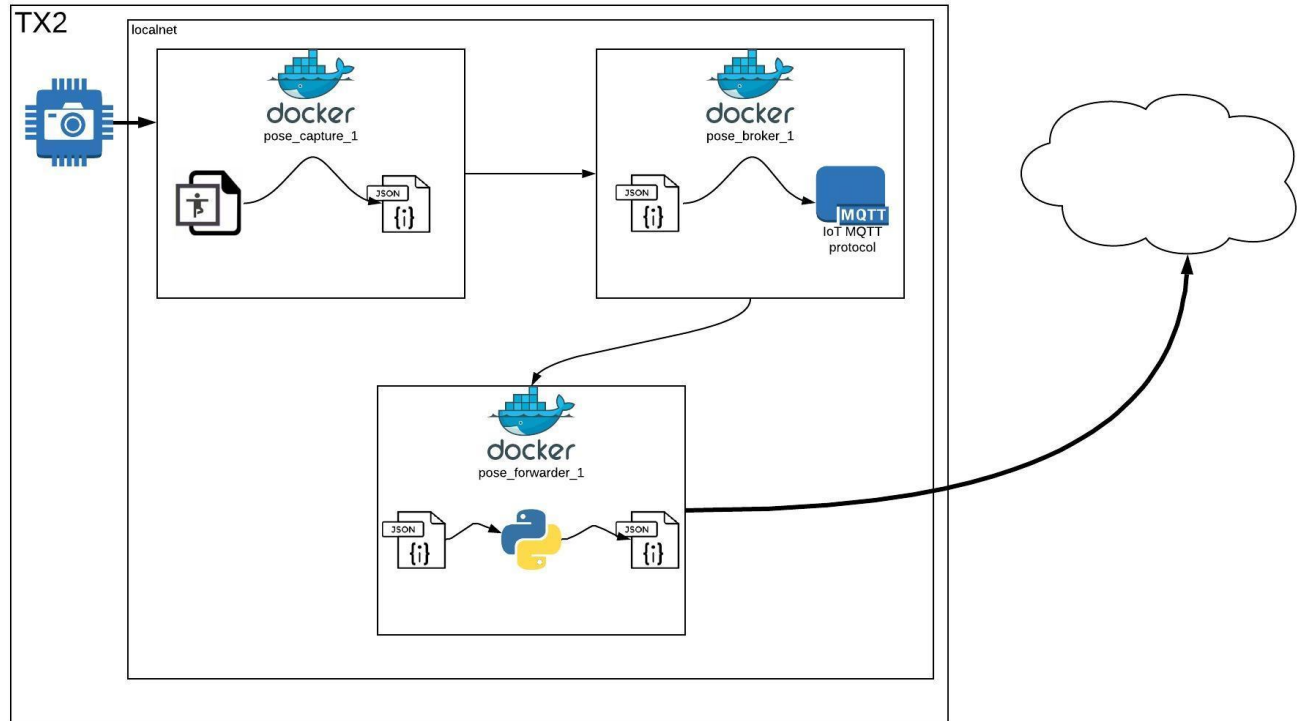
Yoga Pose Classification



CNN

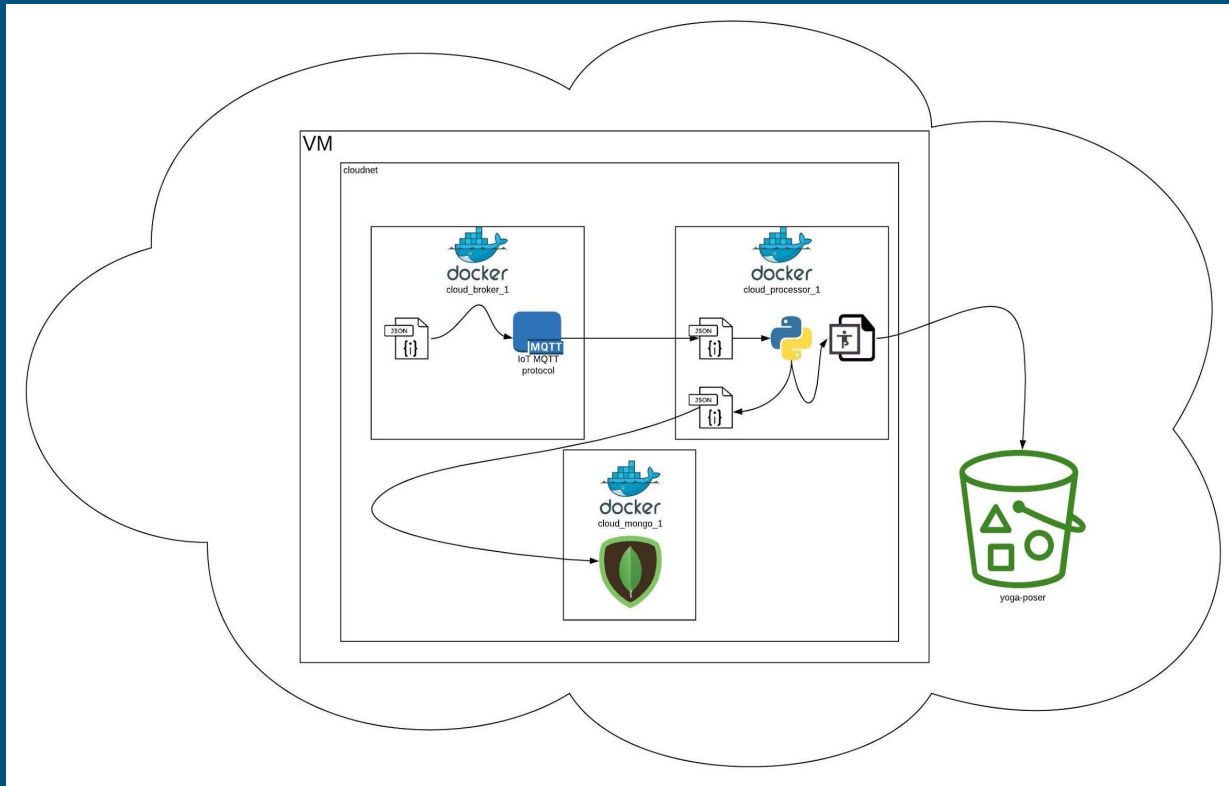
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triangle_pose	0.021
mountain	0.013
tree	0.010
bridge	0.001
childs	0.000
downward_dog	0.000
plank	0.000
seated_forward_bend	0.000

Edge Data Flow



What runs in the cloud

- MQTT Broker
- Image Storage
- Analytics Storage



MQTT messaging

- How are our brokers setup?
- What is the message format?

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Image and Analytics Storage

- How do we store these things?



mongoDB



IBM Cloud
Object Storage

Questions?